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$\langle len(v) \geq 1 \rangle$

```
i=1;
m=0;
while i < len(v) do(
   if v[i] > v[m] then m=i else skip;
   i = i+1;
)
```

$\{0 \leq m < len(v) \wedge \forall j.\ 0 \leq j < len(v) \Rightarrow v[m] \geq v[j]\}$

Invariant: $0 \leq i \leq len(v) \wedge 0 \leq m < len(v) \wedge \forall j.\ 0 \leq j < i \Rightarrow v[m] \geq v[j]$

Termination measure: $len(v) - i$

1. Invariant holds at the beginning of the loop.

$$m=0 \wedge i=1 \wedge len(v) \geq 1 \Rightarrow \underbrace{0 \leq i \leq len(v)}_{i=1 \leq len(v) \geq 1} \wedge \underbrace{0 \leq m < len(v)}_{m=0 < len(v) \geq 1} \wedge \underbrace{\forall j.\ 0 \leq j < i \Rightarrow v[m] \geq v[j]}_{0 \leq j < i = 1 \Rightarrow v[0] \geq v[0]}$$

2. Invariant holds after each iteration.

$$(0 \leq i \leq len(v) \wedge 0 \leq m < len(v) \wedge \forall j.\ 0 \leq j < i \Rightarrow v[m] \geq v[j]) \wedge \overbrace{i < len(v)}^{b} \Rightarrow$$

$$\Big(0 \leq i < len(v) \wedge 0 \leq m < len(v) \wedge \big(v[i] > v[m] \Rightarrow 0 \leq i+1 \leq len(v) \wedge (0 \leq i < len(v) \wedge \forall j.\ (0 \leq j < i+1 \Rightarrow v[i] \geq v[j]))\big)$$
$$\wedge \big(v[i] \leq v[m] \Rightarrow (0 \leq i+1 \leq len(v) \wedge 0 \leq m < len(v) \wedge \forall j.\ (0 \leq j < i+1 \Rightarrow v[m] \geq v[j]))\big) \Big)$$

This means the invariant and the while condition consider the two do branches, either when we update the maximum "m" index when $v[i] > v[m]$ or when we skip, 2nd and 3rd clause respectively. In both cases we end up adding 1 to i, that's why we start with $0 \leq i < len(v)$.

3. Postcondition holds at the end of the loop.

$$\overbrace{(0 \leq i \leq len(v) \wedge 0 \leq m < len(v) \wedge \forall j.\ 0 \leq j < i \Rightarrow v[m] \geq v[j])}^{\text{invariant}} \wedge \overbrace{i \geq len(v)}^{\neg b \downarrow \neg(i < len(v))} \Rightarrow$$

$$\underbrace{0 \leq m < len(v) \wedge \forall j.\ 0 \leq j < len(v) \Rightarrow v[m] \geq v[j]}_{\Psi:\ \text{postcondition}}$$

4. Termination measure is non-negative:

$$(0 \leq i < len(v) \wedge 0 \leq m < len(v) \wedge \forall j.\ 0 \leq j < i \Rightarrow v[m] \geq v[j]) \Rightarrow len(v) - i \geq 0$$

The invariant, which holds at the end of the loop, establishes that $len(v) - i \geq 0$

5. Termination measure decreases after each iteration:

$$(0 \leq i \leq len(v) \wedge 0 \leq m < len(v) \wedge \forall j.\ 0 \leq j < i \Rightarrow v[m] \geq v[j]) \wedge i < len(v) \wedge len(v) - i = k \Rightarrow len(v) - (i+1) < k$$